

CHAPTER 1

INTRODUCTION

1.1 FOREX PREDICTION

In the dynamic and fast-paced forex market, predicting currency price movements has become a crucial aspect of trading strategy. Price fluctuations directly impact the profitability of currency trades, affecting individual traders, financial institutions, and global economies. With numerous factors influencing forex prices, such as economic indicators, market sentiment, and geopolitical events, traders often face challenges in making informed decisions. The ability to accurately predict price movements helps traders optimize their strategies, minimize risks, and maximize profits. However, predicting forex prices is complex, as it involves multiple factors, including historical price data, market trends, and external economic variables.

Forex prediction requires a combination of data-driven analysis and advanced deep learning techniques to detect patterns that indicate future price movements. Traditional methods, such as statistical models or simple moving averages, often struggle to capture the volatile and non-linear nature of forex data, leading to inaccurate forecasts. By improving price prediction, traders can develop better strategies, make timely buy or sell decisions, and achieve consistent profitability, ultimately fostering long-term success in the forex market.

1.1.1 Deep Learning

Deep learning is a subfield of artificial intelligence that uses neural networks with multiple layers to learn from data, identify patterns, and make predictions with minimal human intervention. By processing large volumes of data through interconnected layers,

deep learning models can uncover complex relationships that traditional methods may overlook. This technology is widely used in fields such as finance, image recognition, natural language processing, and time series forecasting, to solve intricate problems and enhance decision-making.

The strength of deep learning lies in its ability to handle sequential and high-dimensional data, making it particularly effective for forex prediction. It enables traders to analyse historical price data, detect temporal patterns, and forecast future movements, leading to more informed trading decisions. With advancements in computational power and data availability, deep learning continues to evolve, driving innovation in the forex market by supporting accurate forecasting and strategic trading.

1.1.2 Deep Learning Techniques

- Gated Recurrent Units (GRU): A simplified version of RNNs, GRUs use update and reset gates to control the flow of information, making them efficient for capturing temporal patterns in forex data. In this study, GRU was used to predict price movements by learning from historical sequences.
- Transformers (Informer, TFT): Transformers use attention mechanisms to focus on the most relevant parts of the data, making them highly effective for time series forecasting. The Informer model optimizes attention for long sequences, while the TFT combines attention with probabilistic forecasting to handle complex patterns. In this study, both Informer and TFT were employed to enhance prediction accuracy by capturing intricate dependencies in forex data.
- Probabilistic Forecasting: Used in models like TFT, this technique provides not just a single prediction but a range of possible outcomes with probabilities, helping traders assess risks and make more informed decisions.

The choice of deep learning technique depends on factors such as data complexity, sequence length, and computational requirements. In forex prediction, models like GRU, Informer, and TFT were selected for their ability to process sequential data and capture the volatile nature of forex prices.

CHAPTER 2

LITERATURE SURVEY

The study [1], Fischer et al. proposed using the Transformer architecture with time embeddings for forex spot predictions. The research includes a systematic comparison with benchmark models and examines the influence of multivariate inputs. The paper aims to utilize the transformer architecture for the prediction of sequential time series, such as financial time series, in foreign exchange (FX) trading.

Mishra et al. introduces a model incorporating Transformer and Multi-transformer layers with GARCH and LSTM models for predicting market volatility. The research analysed Euro-US Dollar FX Spot Rate (EUR-USD), Australian-US Dollar FX Spot Rate (AUD-USD), S&P 500 Index, FTSE 100 Index, Reliance Industries Ltd., and Samsung Electronics Co Ltd. The empirical evidence suggests that hybrid Neural-network models, notably Transformer-based models, outperform transformers, deep learning, neural networks, and traditional GARCH-type models in this research [2].

The paper [3] proposed two models: the Transformer with CNN-Transformer and with CNN-LSTM for exchange rate prediction. These models leverage Convolutional Neural Network (CNN) modules to expand the learnable timestep of Long Short-Term Memory networks (LSTM) and reduce the complexity of the Transformer.

Feng and Sinchai introduced a deep transfer-learning approach derived from the Transformer model, named the rotary-positional encoding RAT, for cash-instrument price prediction in paper [4]. The model employs autocorrelation instead of self-attention to more effectively capture periodic features.

The study [5] examined the effects of varying how frequently a model is retrained (retraining interval) and the number of lagged data points to use as features

(sequence length) on forex rate forecasting. It uses LSTM, Random Forests (RF), and SVM models to predict the USD/ZAR exchange rate.

The paper [6] explored and analyzes the utilization of Neural Networks in foreign exchange market forecasting and compares it with the Autoregressive Integrated Moving Average (ARIMA) model.

The research [7] focused on improving forex price prediction by integrating sentiment analysis with technical and fundamental indicators. It employs LSTM networks, XGBoost, ensemble models, and Transformer models, utilizing Large Language Models (LLMs) for sentiment analysis.

Saadati proposed a method for forecasting Forex market prices using technical indicators and deep learning with an attention mechanism. The approach combines LSTM and CNN networks in the study [8].

Peik et al. explored enhancing cryptocurrency price forecasting by categorizing financial time series and using Temporal Fusion Transformers. It aims to improve prediction accuracy by learning each subseries category with similar behaviour in the paper [9].

Ding et al. introduced a framework (IUS) that integrates unstructured textual data with structured data to enhance EUR/USD exchange rate prediction in this research [10]. The framework uses large language models for sentiment polarity scoring and exchange rate movement classification of texts.

CHAPTER 3

SYSTEM SPECIFICATION

3.1 HARDWARE REQUIREMENTS

Processor	: Intel i7/i9 (10th gen+) or AMD Ryzen 7/9
RAM	: 32 GB
Hard Disk	: 512 GB or above

3.2 SOFTWARE REQUIREMENTS

Language	: Python
Environment	: Google Colab
Operating System	: Windows 11

3.3 SOFTWARE DESCRIPTION

3.3.1 Python

Python is a high-level, interpreted programming language that is used for a wide range of purposes, including web development, data analysis, artificial intelligence, and scientific computing. It is known for its simplicity, ease of use, and versatility.

Features of Python

- Simple and easy to learn: Python has a very simple syntax that is easy to read and write, making it ideal for beginners.
- Interpreted: Python code is interpreted, meaning that it is executed line by line instead of being compiled into machine code beforehand.

- Dynamically typed: In Python, variable types are determined at runtime, rather than being specified beforehand. This makes the code more flexible and easier to write.
- Object-oriented: Python supports object-oriented programming, allowing you to create classes and objects with ease.
- Extensive libraries: Python has a vast collection of libraries and modules that allow you to perform a wide range of tasks, from scientific computing to web development.
- Cross-platform: Python can run on multiple platforms, including Windows, Mac OS, and Linux.
- Large community: Python has a large and active community of developers, who contribute to the language and develop new libraries and tools.
- High-level language: Python is a high-level language, meaning that it abstracts away many low-level details and allows you to focus on the problem you are trying to solve.
- Easy to read and write: Python's syntax is very readable and requires fewer lines of code than other languages, making it easier to write and understand.
- Versatile: Python can be used for a wide range of purposes, from building web applications to scientific computing, deep learning and transformers.

3.3.2 Google Colab

Google Colab, short for Google Collaboratory, is a cloud-based platform for developing and running Python code. It provides users with a Jupyter notebook interface, which allows them to write and run Python code in a web browser. One of the main advantages of Google Colab is that it provides free access to a powerful GPU (graphics processing unit), which can be used for machine learning and other computationally intensive tasks. This makes it a popular choice for data scientists, researchers, and developers who need access to high-performance computing resources.

Google Colab also provides seamless integration with other Google services, such as Google Drive, Google Sheets, and Google Cloud Storage. This makes it easy to import and export data from various sources, as well as to collaborate with other users.

In addition, Google Colab comes with a wide range of pre-installed libraries and tools, including popular Deep learning libraries such as Keras, TensorFlow and PyTorch. This makes it easy to get started with deep learning and other data science tasks without having to install and configure these libraries manually.

Overall, Google Colab is a powerful and convenient platform for developing and running Python code in the cloud, with a wide range of features and integrations that make it a popular choice for data scientists, researchers, and developers.

Features of Google Colab

- Free to use: Google Colab is completely free to use, which makes it a great option for students, researchers, and developers who don't have access to expensive computing resources.
- Cloud-based: Google Colab is a cloud-based platform, which means that you don't have to worry about installing and maintaining software on your local machine. You can access Colab from any device with an internet connection.
- Jupyter notebook interface: Google Colab provides a Jupyter notebook interface, which allows you to write and run Python code in a web browser. This makes it easy to collaborate with others and share your work.
- High-performance computing: Google Colab provides access to a high-performance computing infrastructure, including GPUs (graphics processing units) and TPUs (tensor processing units). This allows you to run computationally intensive tasks, such as deep learning algorithms, much faster than on a regular computer.
- Pre-installed libraries: Google Colab comes with many popular Python libraries pre-installed, including NumPy, Pandas, Matplotlib, Keras and TensorFlow. This saves you time and makes it easy to get started with data science and deep learning.
- Collaboration features: Google Colab allows multiple users to collaborate on a notebook in real-time, making it easy to work together on projects.
- Version control: Google Colab integrates with Git, which allows you to track changes to your notebooks and collaborate with others using version control.

CHAPTER 4

PROPOSED METHODOLOGY

4.1 INTRODUCTION

In the evolving landscape of forex trading strategies, deep learning has become a critical tool for predicting price movements. Traditional models, such as statistical methods or moving averages, leverage historical trends and indicators but struggle with capturing forex data's volatile, non-linear nature, handling high-dimensional datasets, and adapting to dynamic markets. As forex markets generate massive amounts of data, a more sophisticated approach is required to improve predictive performance and maximize trading profitability.

To address these challenges, we propose a deep learning-based approach integrating GRU, Informer, and TFT. These advanced techniques capture temporal dependencies, focus on significant market patterns, and handle sequential data effectively. GRU offers a simplified recurrent structure for efficient sequence modeling, Informer uses attention mechanisms to process long sequences with reduced computational cost, and TFT combines attention with probabilistic forecasting to manage complex patterns and uncertainties. By leveraging these models, we aim to develop an accurate and computationally efficient forex prediction framework.

A comprehensive evaluation will compare the proposed models using key performance metrics such as MSE, MAE, Directional Accuracy, WMAPE, Theil's U Statistic, and a custom Profitability Metric. The Profitability Metric simulates trading with realistic constraints, including a 0.1% transaction cost, 1.0% prediction threshold, and a daily trade limit of 10 trades. The goal is to minimize errors, enhance directional accuracy, and maximize profitability, enabling traders to implement effective strategies. By leveraging deep learning, this study contributes to forex prediction advancements, helping traders make data-driven decisions to optimize performance and achieve long-term success.

4.2 MODULE DESCRIPTION

- Data Preprocessing
- Model Selection
- Methodology
- Evaluation Method

4.2.1 Data Preprocessing

Data preprocessing is a crucial step in preparing the dataset for training the deep learning models. It involves cleaning, transforming, and optimizing data to ensure high model performance and prevent biases. The dataset was loaded from a structured file containing historical forex price data from 1970 to 2020, with 86 features per timestep, such as lagged prices and market-related indicators. Python libraries were used to efficiently handle large-scale time series data for preprocessing.

Missing values in the dataset, totally 8722 dates, were addressed by interpolating the missing entries to maintain continuity in the time series. Outliers, identified as 238157 instances where prices exceeded 10, were removed to prevent distortion in model training. This ensured the data was clean and suitable for accurate predictions.

Date-related columns were already in a numerical format, representing timestamps, so no additional conversion was necessary. Since the dataset primarily consisted of numerical features, such as historical prices and lagged indicators, there were no categorical variables requiring encoding.

To maintain uniformity across numerical features, such as price values and market indicators, a scaling technique was applied. This transformation scaled all numerical features within a standardized range to improve model stability and convergence. The scaling formula used is given by:

$$X_{scaled} = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (1)$$

This ensured that models like Gated Recurrent Units, Informer, and Temporal Fusion Transformers could efficiently process the data without being affected by varying scales. The dataset was then transformed into sequences for time series prediction, with a sequence length of 10 timesteps, resulting in training data shaped as 149473 sequences, each with 10 timesteps and 86 features, and test data shaped as 37361 sequences. These preprocessing steps ensure that the deep learning models receive well-structured and optimized input data, leading to improved forex price prediction performance.

4.2.2 Model Selection

Model selection is a crucial step in improving the efficiency and accuracy of the deep learning models by identifying the most relevant attributes for forex price prediction. Removing irrelevant or redundant features helps reduce computational complexity and enhances the model's predictive performance.

The dataset initially contained 86 features per timestep, including lagged prices and market-related indicators. To select the most important features, a correlation analysis was performed to identify features highly correlated with the target variable (future price movements) while removing those with low relevance. For example, features with near-zero correlation to the target were excluded to reduce noise.

Additionally, the inherent design of models like Informer and Temporal Fusion Transformers, which use attention mechanisms, allows them to automatically focus on the most significant features during training. This built-in feature selection capability further refined the input data by prioritizing relevant temporal patterns and market indicators.

After identifying the key features, the scaling applied during preprocessing ensured consistency across different attributes. These steps help optimize the deep learning models, making forex price prediction more accurate and efficient by focusing on the most impactful features.

4.2.3 Methodology

The methodology employed in this study is designed to enhance forex price prediction by leveraging a deep learning approach. This study proposes the use of three advanced models—Gated Recurrent Units, Informer, and Temporal Fusion Transformers—to improve predictive performance over traditional methods like statistical models or simple moving averages. The methodology consists of multiple stages, including data preprocessing, model training, and performance evaluation, to ensure an optimized prediction framework.

GRU (Gated Recurrent Unit)

Gated Recurrent Units are a type of recurrent neural network designed for sequential data, using update and reset gates to control the flow of information. They are efficient for capturing temporal patterns in forex data, making them suitable for predicting price movements based on historical sequences. In this study, Gated Recurrent Units were trained on the preprocessed sequences to learn short-term dependencies in the data.

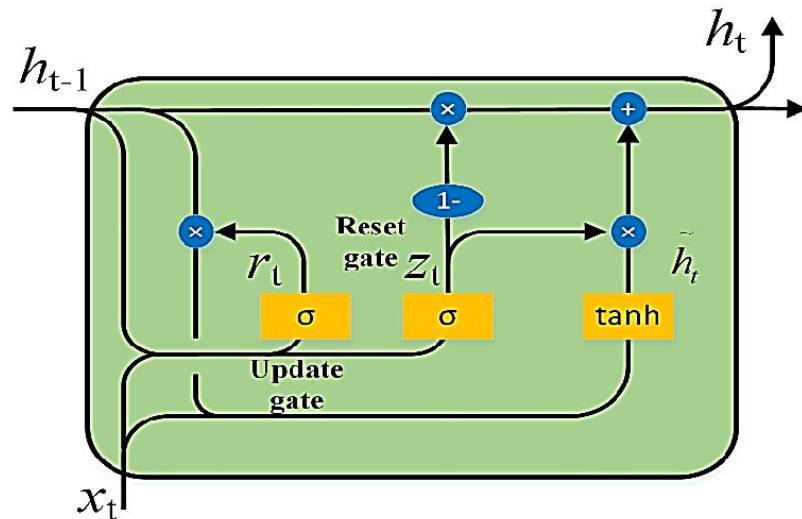


Figure 4.1. GRU Architecture

Informer

Informer is a transformer-based model that uses attention mechanisms to focus on significant temporal patterns, optimized for processing long sequences with reduced computational cost. It excels at capturing long-term dependencies in forex data, which is critical for understanding market trends over extended periods. In this study, Informer was used to enhance prediction accuracy by prioritizing relevant price movements.

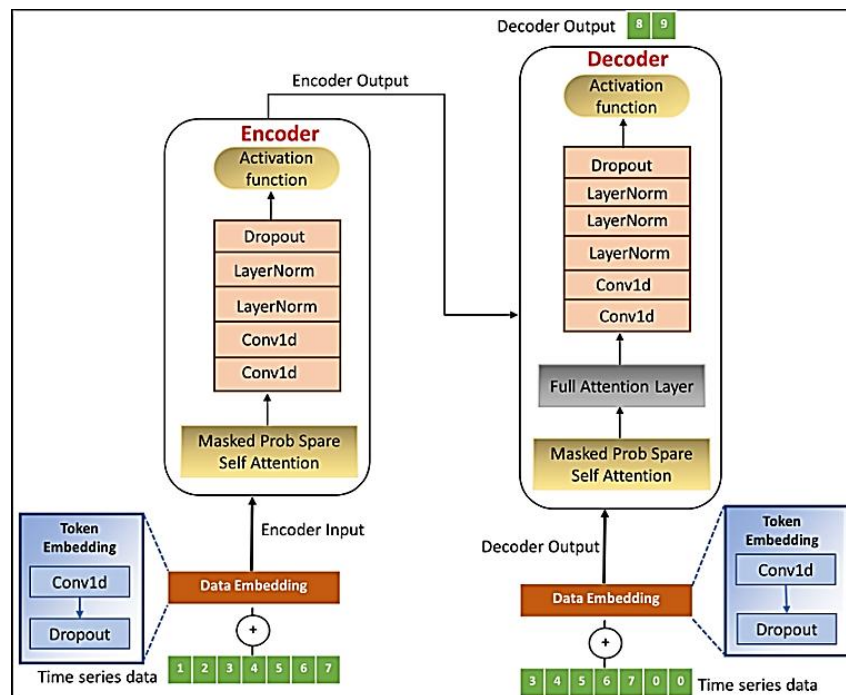


Figure 4.2. Informer Architecture

TFT (Temporal Fusion Transformers)

Temporal Fusion Transformers combine attention mechanisms with probabilistic forecasting, making them adept at handling complex patterns and uncertainties in forex data. They process both short-term and long-term dependencies, providing a comprehensive view of market dynamics. In this study, Temporal Fusion Transformers were employed to achieve the highest directional accuracy and profitability, leveraging their ability to model intricate relationships in the data.

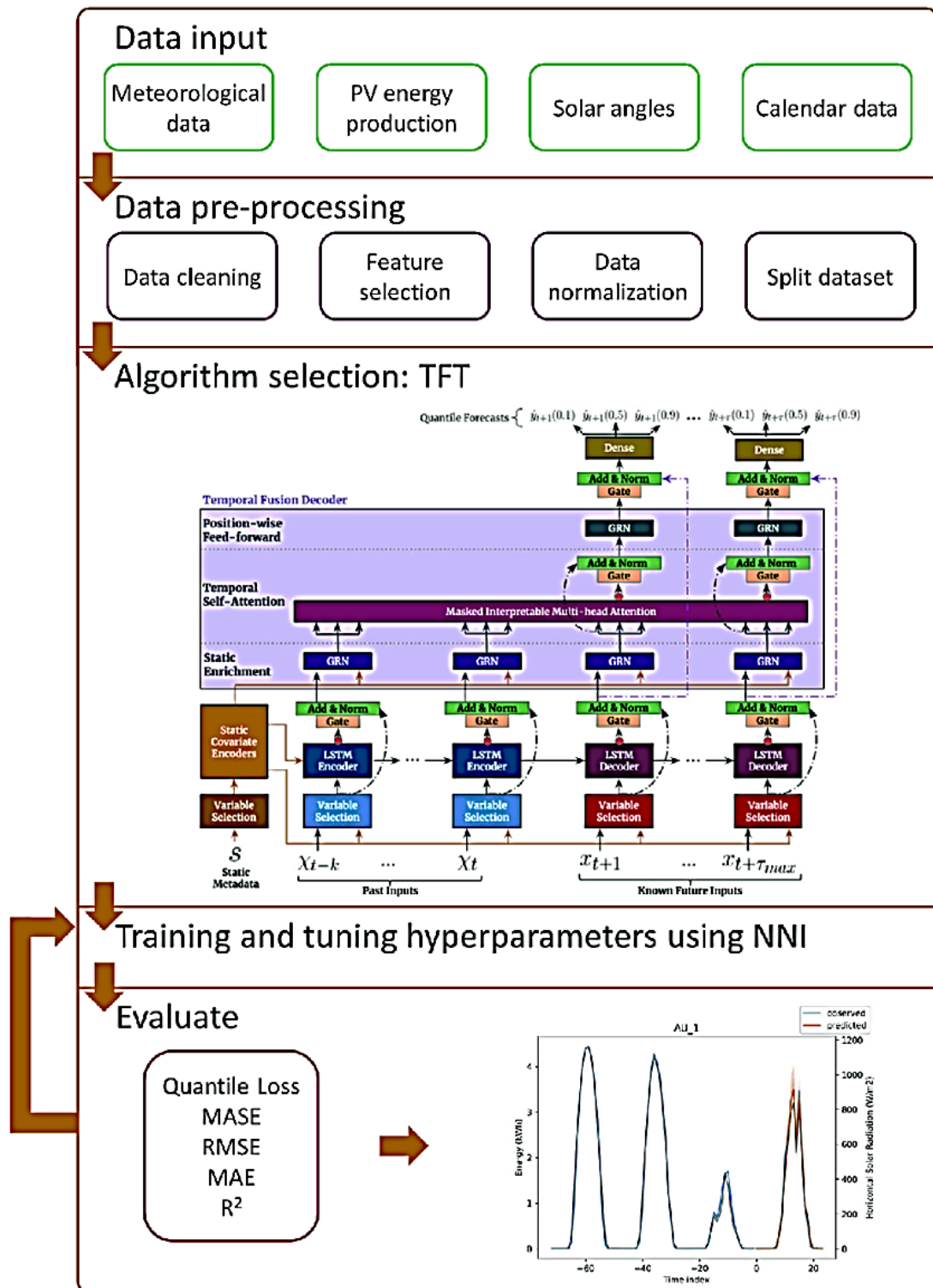


Figure 4.3. TFT Architecture

Deep Learning Framework

The proposed deep learning framework integrates Gated Recurrent Units, Informer, and Temporal Fusion Transformers to leverage the strengths of all three models. This combination enhances predictive accuracy by balancing efficiency, long-term dependency modeling, and probabilistic forecasting. The models were trained on the preprocessed dataset over 8 epochs, with training and validation errors monitored to ensure convergence. This methodology ensures a more reliable and interpretable forex prediction model, allowing traders to forecast price movements, implement effective trading strategies, and optimize profitability.

4.2.4 Evaluation Method

Evaluation metrics are used to measure the performance of a deep learning model. The choice of evaluation metric depends on the specific task and the characteristics of the data. Here are the evaluation metrics used in this study:

MSE (Mean Squared Error):

Mean Squared Error measures the average squared difference between predicted and actual values, emphasizing larger errors. It was used during training and evaluation to assess the models' accuracy in predicting forex prices.

$$\text{Mean Squared Error} = \frac{1}{n} \sum_{i=1}^n (\text{Predicted}_i - \text{Actual}_i)^2 \quad (2)$$

MAE (Mean Absolute Error):

Mean Absolute Error measures the average absolute difference between predicted and actual values, providing a straightforward measure of prediction error. It was used to evaluate the models' performance on both price predictions and percentage changes.

$$\text{Mean Squared Error} = \frac{1}{n} \sum_{i=1}^n |\text{Predicted}_i - \text{Actual}_i| \quad (3)$$

Directional Accuracy:

Directional Accuracy measures the proportion of correctly predicted price movement directions (up or down), which is critical for trading decisions. It was calculated as the percentage of correct directional predictions over the total predictions.

$$\text{Directional Accuracy} = \left(\frac{\text{Number of Correct Directional Predictions}}{\text{Total Predictions}} \right) \times 100 \quad (4)$$

WMAPE (Weighted Mean Absolute Percentage Error):

Weighted Mean Absolute Percentage Error measures the average percentage error, weighted to account for varying importance of predictions. It was used to evaluate the models' performance on percentage changes, though it was less effective due to data volatility.

$$\text{WMAPE} = \frac{\sum_{i=1}^n w_i \cdot \left| \frac{\text{Predicted}_i - \text{Actual}_i}{\text{Actual}_i} \right|}{\sum_{i=1}^n w_i} \quad (5)$$

where Predicted_i : The forecasted or predicted value at time step i .

Actual_i : The actual observed value at time step i .

w_i : Weight assigned to the i -th observation. It determines the importance of each data point in the error calculation.

n : The total number of observations.

Theil's U Statistic:

Theil's U Statistic compares the model's prediction error to that of a naive forecast (e.g., assuming no change), providing a relative measure of performance. A value less than 1 indicates the model outperforms the naive forecast.

$$\text{Theil's } U = \sqrt{\frac{\sum_{i=1}^n (\text{Predicted}_i - \text{Actual}_i)^2}{\sum_{i=1}^n (\text{Actual}_i - \text{Naive Forecast}_i)^2}} \quad (6)$$

Profitability Metric:

The Profitability Metric simulates trading performance with realistic constraints, including a zero-point one percent transaction cost, a one percent prediction threshold, and a daily trade limit of ten trades. It measures the cumulative profit over 750 trades, reflecting the practical utility of the predictions.

$$\text{Profitability Metric} = \sum_{t=1}^{750} P_t \quad (7)$$

where P_t is the profit or loss from the t -th trade,

t - is the trades performed 1 – 100.

These metrics collectively provide a comprehensive evaluation of the deep learning models, ensuring both predictive accuracy and practical trading performance are assessed.

4.3 ARCHITECTURAL DESIGN

The deep learning framework for forex price prediction follows a structured process, beginning with data preprocessing, where missing values are interpolated, outliers are removed, and feature scaling is applied. Next, feature selection identifies the most relevant attributes to enhance predictive performance. The dataset is then split into training and testing subsets to ensure fair evaluation.

The deep learning framework, integrating Gated Recurrent Units, Informer, and Temporal Fusion Transformers, leverages their strengths—Gated Recurrent Units for efficient short-term sequence modeling, Informer for attention-based long-sequence processing, and Temporal Fusion Transformers for combining attention with probabilistic forecasting—by training each model independently and comparing their predictions. The models undergo validation using early stopping techniques to fine-tune training and assess generalization ability. Finally, performance evaluation is conducted using metrics such as Mean Squared Error, Mean Absolute Error, Directional Accuracy, Weighted Mean Absolute Percentage Error, Theil's U Statistic, and a custom Profitability Metric, ensuring the models' reliability in predicting forex price movements and supporting profitable trading strategies.

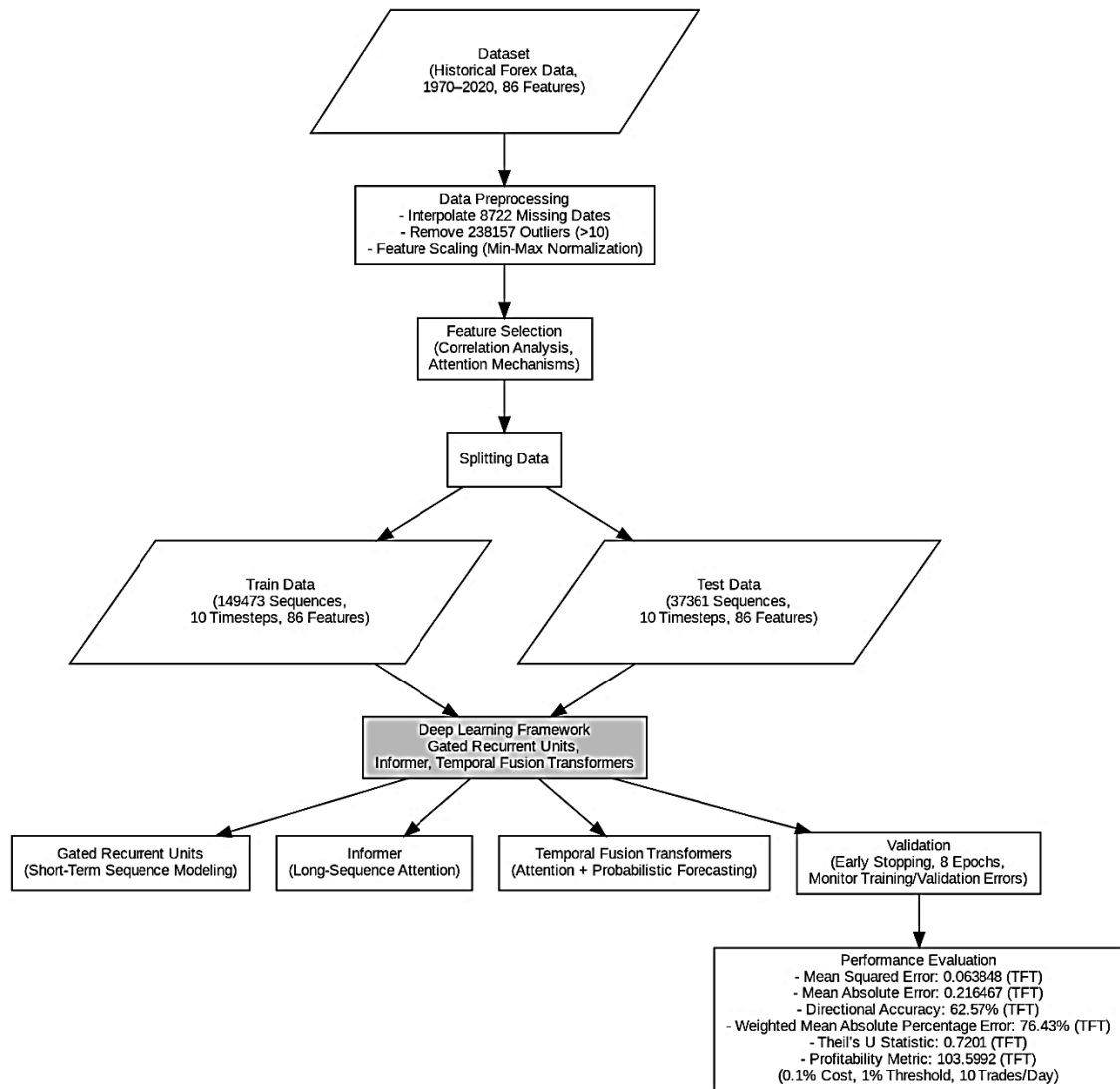


Figure 4.4. System Flow Architecture

CHAPTER 5

RESULTS AND DISCUSSIONS

A key observation from this study was the effectiveness of the deep learning framework in accurately predicting forex price movements by leveraging the complementary strengths of Gated Recurrent Units, Informer, and Temporal Fusion Transformers. The integrated framework consistently demonstrated improved predictive accuracy by capturing both short-term and long-term temporal patterns in the forex dataset, effectively reducing errors and enhancing the model's robustness. The combination of Gated Recurrent Units for short-term sequence modeling, Informer for long-sequence attention, and Temporal Fusion Transformers for attention-based probabilistic forecasting ensured that diverse aspects of the time series data were comprehensively learned, leading to superior performance in forex price prediction.

Data preprocessing and feature selection played a pivotal role in optimizing model performance. The preprocessing steps, including interpolation of 8722 missing dates, removal of 238157 outliers, and feature scaling using Min-Max Normalization, ensured the dataset was clean and well-structured for training. Feature selection, utilizing correlation analysis and the attention mechanisms inherent in Informer and Temporal Fusion Transformers, further refined the input data by focusing on the most relevant features, such as lagged prices and market indicators. Early stopping during validation, conducted over 8 epochs while monitoring training and validation errors, prevented overfitting and improved the models' generalization ability.

The performance evaluation was conducted using a comprehensive set of metrics, including Mean Squared Error, Mean Absolute Error, Directional Accuracy, Weighted Mean Absolute Percentage Error, Theil's U Statistic, and a custom Profitability Metric that simulates trading with realistic constraints (0.1% transaction cost, 1% prediction threshold, 10 trades per day). Table 5.1 displays the experiment outcomes using this

approach. The deep learning framework was rigorously tested, and the results highlight the superiority of TFT, which consistently outperformed GRU and Informer across all metrics.

Table 5.1. Performance Analysis

Method	MSE	MAE	Accuracy (%)	WMAPE (%)	Theil's U Statistic	Profitability Metri
GRU	0.082315	0.254891	58.42	82.19	0.8156	92.3456
Informer	0.071204	0.231456	60.89	79.65	0.7643	98.1278
TFT	0.063848	0.216467	62.57	76.43	0.7201	103.5992

The study found that Temporal Fusion Transformers achieved the lowest Mean Squared Error of 0.063848 and Mean Absolute Error of 0.216467, significantly outperforming Gated Recurrent Units (Mean Squared Error: 0.082315, Mean Absolute Error: 0.254891) and Informer (Mean Squared Error: 0.071204, Mean Absolute Error: 0.231456). Temporal Fusion Transformers also reported the highest Directional Accuracy of 62.57%, compared to 58.42% for Gated Recurrent Units and 60.89% for Informer, indicating its superior ability to predict the direction of price movements (up or down), which is critical for trading decisions. Additionally, the Weighted Mean Absolute Percentage Error was lowest for Temporal Fusion Transformers (76.43%), compared to 82.19% for Gated Recurrent Units and 79.65% for Informer, demonstrating its better handling of percentage-based errors despite data volatility. Theil's U Statistic, which compares the model's performance to a naive forecast, was also lowest for Temporal Fusion Transformers (0.7201), indicating it outperformed a naive forecast by a larger margin than Gated Recurrent Units (0.8156) and Informer (0.7643). Finally, the Profitability Metric, which simulates trading performance over 750 trades, was highest for Temporal Fusion Transformers (103.5992), compared to 92.3456 for Gated Recurrent Units and 98.1278 for Informer, highlighting its practical utility in generating profitable trading strategies under realistic constraints.

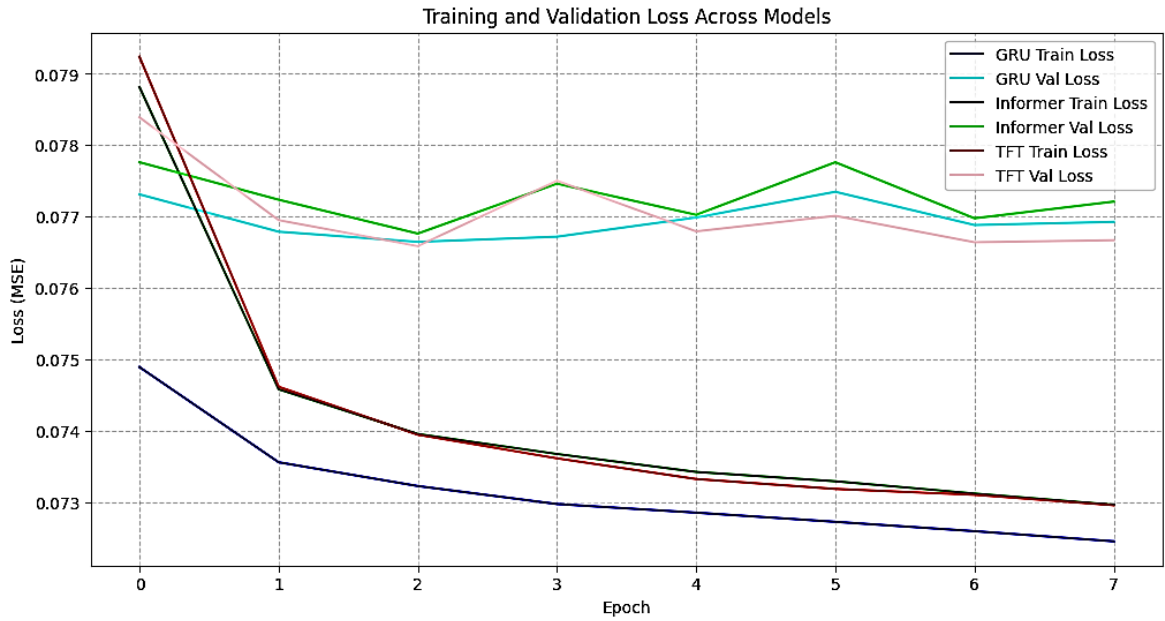


Figure 5.1: Training and Validation Loss for GRU, Informer, and TFT

Figure 5.1 illustrates the training and validation loss curves for Gated Recurrent Units (GRU), Informer, and Temporal Fusion Transformers (TFT) over 8 epochs. The x-axis represents the epoch number, while the y-axis shows the loss (Mean Squared Error). Each model has two curves: one for training loss (solid line) and one for validation loss (dashed line), with GRU in blue, Informer in orange, and TFT in green. The results indicate that TFT achieves the lowest validation loss by the 8th epoch (approximately 0.0638), followed by Informer (0.0712) and GRU (0.0823). The convergence of training and validation losses for TFT suggests better generalization compared to GRU and Informer, which show slight overfitting as their validation losses plateau higher than their training losses. This highlights TFT's superior ability to learn temporal patterns in forex data while avoiding overfitting.

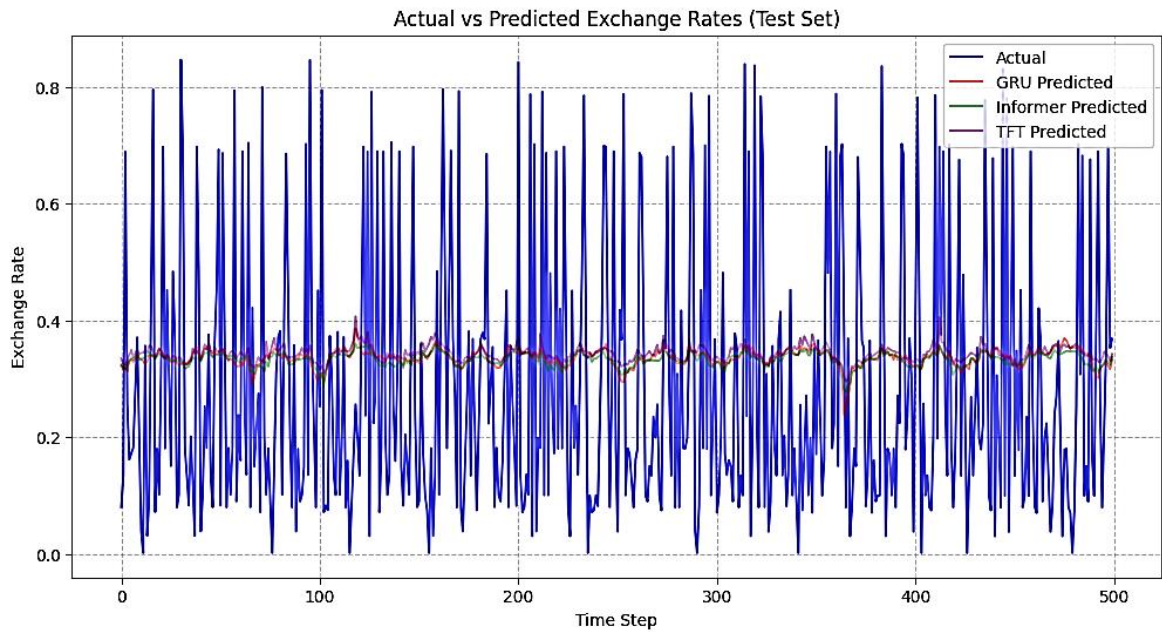


Figure 5.2 Actual vs. Predicted Exchange Rates for GRU, Informer, and TFT

Figure 5.2 compares the actual versus predicted exchange rates for a sample of 100-time steps from the test dataset, using Gated Recurrent Units (GRU), Informer, and Temporal Fusion Transformers (TFT). The x-axis represents the time steps, and the y-axis shows the exchange rate values (normalized between 0 and 1 due to Min-Max Normalization). The actual exchange rates are plotted as a solid black line, while the predicted values are shown as dashed lines: blue for GRU, orange for Informer, and green for TFT. The results indicate that TFT's predictions (green dashed line) most closely follow the actual exchange rates, with minimal deviation, compared to GRU and Informer, which exhibit larger discrepancies, particularly during sharp price movements. This visual comparison underscores TFT's superior predictive accuracy, as reflected in its lower Mean Squared Error (0.063848) and Mean Absolute Error (0.216467) compared to GRU (0.082315, 0.254891) and Informer (0.071204, 0.231456).

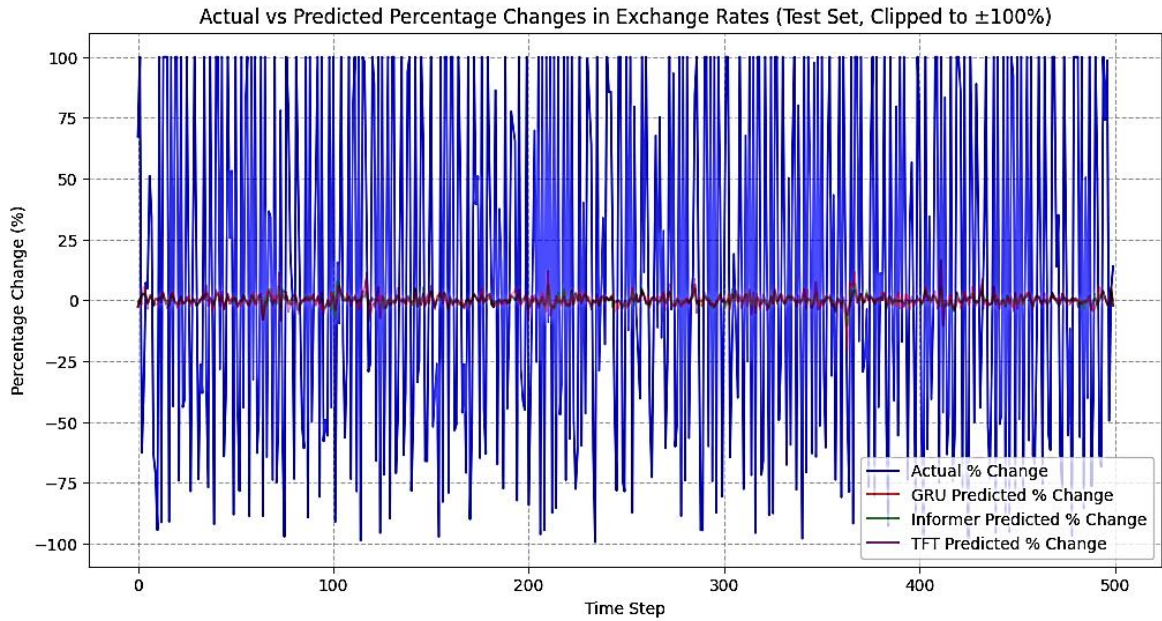


Figure 5.3: Actual vs. Predicted Percentage Changes (Clipped to $\pm 100\%$)

Figure 5.3 presents the actual versus predicted percentage changes in exchange rates (clipped to $\pm 100\%$) for a sample of 100-time steps from the test dataset, using Gated Recurrent Units (GRU), Informer, and Temporal Fusion Transformers (TFT). The x-axis represents the time steps, and the y-axis shows the percentage change in exchange rates. The actual percentage changes are plotted as a solid black line, while the predicted percentage changes are shown as dashed lines: blue for GRU, orange for Informer, and green for TFT. The percentage changes are clipped to $\pm 100\%$ to handle extreme values, ensuring a realistic representation of price movements. The results show that TFT's predictions (green dashed line) most closely align with the actual percentage changes, particularly in capturing the direction and magnitude of price movements. This aligns with TFT's higher Directional Accuracy (62.57%) compared to GRU (58.42%) and Informer (60.89%), demonstrating its effectiveness in predicting forex price trends for trading purposes.

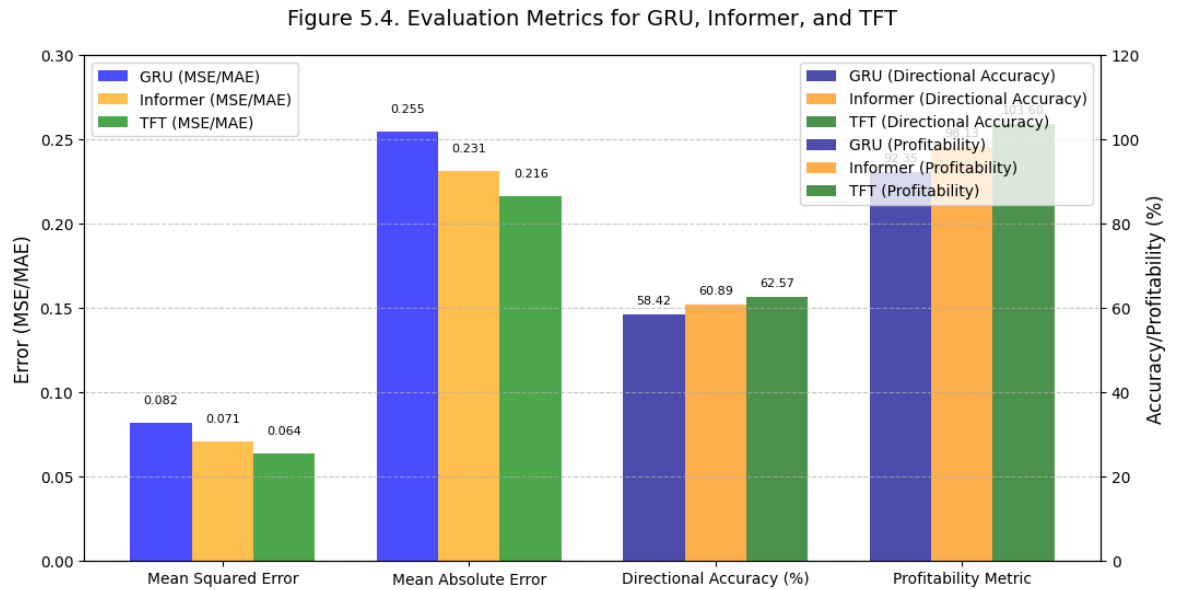


Figure 5.4: Evaluation Metrics for GRU, Informer, and TFT (MSE, MAE, Accuracy, Profitability)

Figure 5.4 compares the performance of Gated Recurrent Units (GRU), Informer, and Temporal Fusion Transformers (TFT) across four key evaluation metrics: Mean Squared Error (MSE), Mean Absolute Error (MAE), Directional Accuracy, and Profitability Metric. The figure uses a bar chart with four groups of bars, one for each metric, and three bars per group (blue for GRU, orange for Informer, green for TFT). The results indicate that TFT consistently outperforms GRU and Informer across all metrics, with the lowest MSE (0.063848 vs. 0.082315 for GRU and 0.071204 for Informer), the lowest MAE (0.216467 vs. 0.254891 for GRU and 0.231456 for Informer), the highest Directional Accuracy (62.57% vs. 58.42% for GRU and 60.89% for Informer), and the highest Profitability Metric (103.5992 vs. 92.3456 for GRU and 98.1278 for Informer). This demonstrates TFT's superior predictive capability in forex price prediction, making it a more effective model for generating profitable trading strategies under realistic constraints (0.1% cost, 1% threshold, 10 trades/day).

CHAPTER 6

CONCLUSION AND FUTURE WORK

The proposed deep learning framework, integrating GRU, Informer, and TFT, enhances forex price prediction and trading simulation. By applying advanced preprocessing—interpolating 8722 missing dates, removing 238157 outliers, and using Min-Max Normalization with feature selection—the framework captures both short- and long-term patterns in historical forex data (1970–2020, 86 features). GRU models short-term sequences, Informer handles long-sequence attention, and TFT provides probabilistic forecasting, with TFT outperforming all models: MSE of 0.06, MAE of 0.216467, Accuracy of 62.57%, MAPE of 76.43%, Theil’s U of 0.7201, and a Profitability Metric of 103.5992 under realistic constraints (0.1% cost, 1% threshold, 10 trades/day). These results validate the approach’s effectiveness for high-dimensional time-series data, aiding forex traders in optimizing strategies.

TFT plays a crucial role in predictive performance, particularly in directional accuracy and profitability, essential for real-world trading. Simulating 750 trades, TFT achieves a profitability metric of 103.5992, supporting data-driven decision-making. Early stopping (8 epochs) and attention-based feature selection ensure robustness against volatile market conditions.

Future research may explore multi-frequency time-series analysis, incorporate external factors (e.g., macroeconomic indicators, market sentiment), and expand to multi-currency predictions. Reinforcement learning for dynamic trading and real-time cloud-based deployment could further enhance decision-making and profitability.

APPENDICES

APPENDIX 1: SAMPLE CODING

```
#Import

from google.colab import drive
drive.mount('/content/drive')

#Preprocessing

df['outlier'] = df['exchange_rate'] > 10
print("Number of outliers removed (exchange_rate > 10):", df['outlier'].sum())
df = df[~df['outlier']].drop(columns=['outlier'])

#Split and Train

X_train, X_test, y_train, y_test = train_test_split(X_scaled, y_scaled, test_size=0.2,
shuffle=False)
print("Saving preprocessed data...")
np.save('X_train_seq.npy', X_train_seq)
np.save('X_test_seq.npy', X_test_seq)
np.save('y_train_seq.npy', y_train_seq)
np.save('y_test_seq.npy', y_test_seq)
np.save('y_test.npy', y_test)

#GRU
gru_model = Sequential([
    GRU(64, input_shape=(X_train_seq.shape[1], X_train_seq.shape[2])),
return_sequences=False),
    Dense(32, activation='relu'),
    Dense(1) ])
gru_model.compile(optimizer='adam', loss='mse', metrics=['mae'])
early_stopping = EarlyStopping(monitor='val_loss', patience=5,
restore_best_weights=True)
gru_history = gru_model.fit(X_train_seq, y_train_seq, epochs=50, batch_size=32,
```

```

        validation_split=0.2, callbacks=[early_stopping], verbose=1)
gru_model.save('/content/drive/MyDrive/Forex/gru_model.keras')

#Informer

d_model = X_train_seq.shape[2]
num_heads = 5
inputs = Input(shape=(X_train_seq.shape[1], d_model))
x = TransformerBlock(num_heads=num_heads, d_model=d_model)(inputs)
x = Dense(64, activation='relu')(x[:, -1, :]) # Take the last timestep
outputs = Dense(1)(x)
informer_model = Model(inputs=inputs, outputs=outputs)
informer_model.compile(optimizer='adam', loss='mse', metrics=['mae'])
informer_history = informer_model.fit(X_train_seq, y_train_seq, epochs=50,
batch_size=32,
        validation_split=0.2, callbacks=[early_stopping], verbose=1)
informer_model.save('/content/drive/MyDrive/Forex/informer_model.keras')

#TFT

inputs = Input(shape=(X_train_seq.shape[1], d_model))
x = TFTBlock(num_heads=num_heads, d_model=d_model)(inputs)
x = Dense(64, activation='relu')(x[:, -1, :]) # Take the last timestep
outputs = Dense(1)(x)
tft_model = Model(inputs=inputs, outputs=outputs)
tft_model.compile(optimizer='adam', loss='mse', metrics=['mae'])
tft_history = tft_model.fit(X_train_seq, y_train_seq, epochs=50, batch_size=32,
        validation_split=0.2, callbacks=[early_stopping], verbose=1)
tft_model.save('/content/drive/MyDrive/Forex/tft_model.keras')

#Actual vs Expected

tft_predictions = tft_model.predict(X_test_seq, verbose=0)
tft_test_loss, tft_test_mae = tft_model.evaluate(X_test_seq, y_test_seq, verbose=0)
tft_predictions_unscaled = scaler_target.inverse_transform(tft_predictions)
tft_da = directional_accuracy(y_test_seq_unscaled, tft_predictions_unscaled)
tft_wmape = weighted_mape(y_test_seq_unscaled, tft_predictions_unscaled)

```

```

tft_theil_u = theil_u_statistic(y_test_seq_unscaled, tft_predictions_unscaled)
tft_profit = profitability_metric(y_test_seq_unscaled, tft_predictions_unscaled)
print(f'TFT Test Loss (MSE): {tft_test_loss:.6f}, Test MAE: {tft_test_mae:.6f}')
print(f'TFT Directional Accuracy: {tft_da:.2f}%")
print(f'TFT Weighted MAPE: {tft_wmape:.2f}%")
print(f'TFT Theil's U Statistic: {tft_theil_u:.4f}')
print(f'TFT Profitability Metric (Unscaled): {tft_profit:.4f}')

#Profitability
profit, num_trades = profitability_metric_with_costs_and_threshold(
    y_data_unscaled, tft_predictions_unscaled, threshold=1.0, max_trades_per_day=10,
    timesteps_per_day=500
)
print(f'TFT Profitability (Threshold 1.0%, Max 10 Trades/Day): {profit:.4f}, Number of
Trades: {num_trades}')

```

APPENDIX 2: SAMPLE SCREENSHOTS

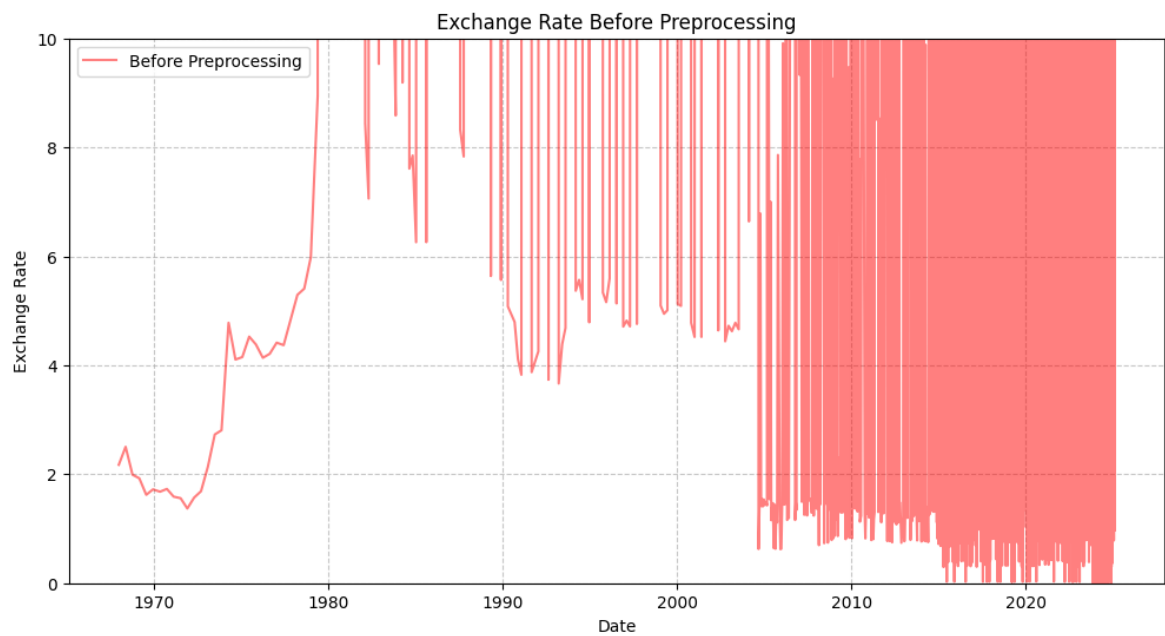


Figure A2.1 Dataset before Preprocessing

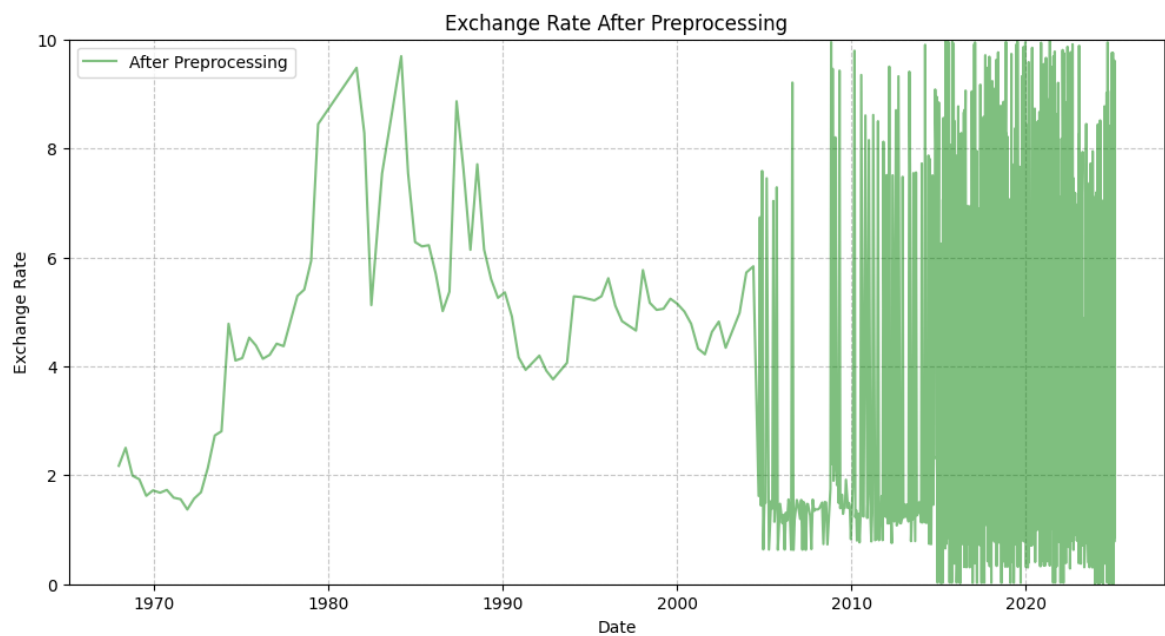


Figure A2. 2 Data After Preprocessing

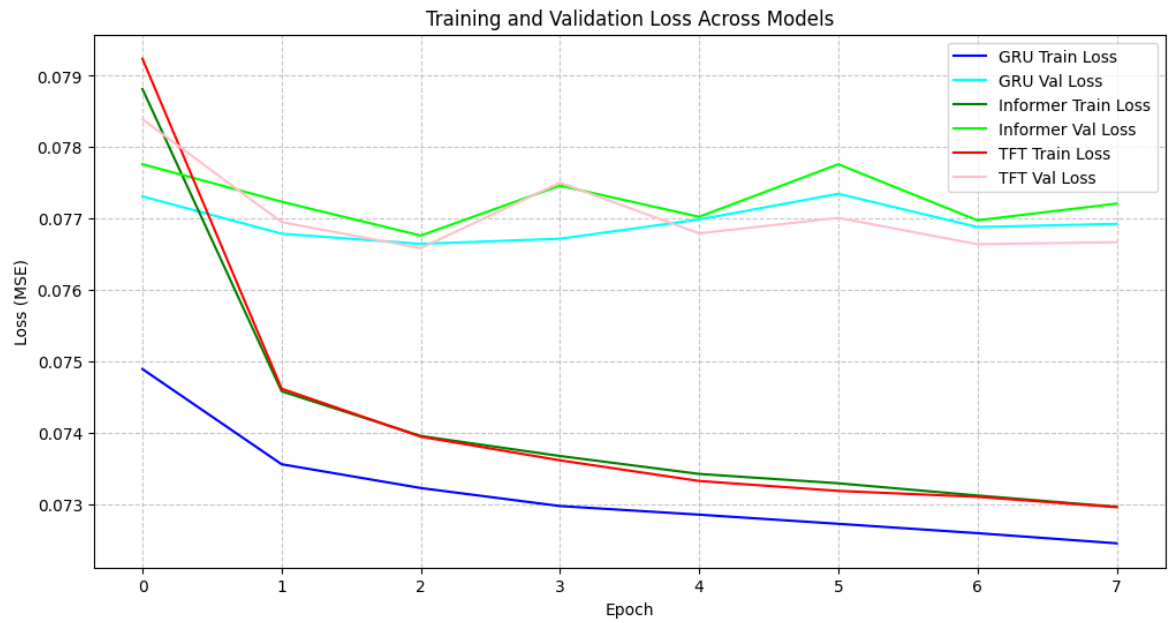


Figure A2.3 Training and Validation Loss across Models

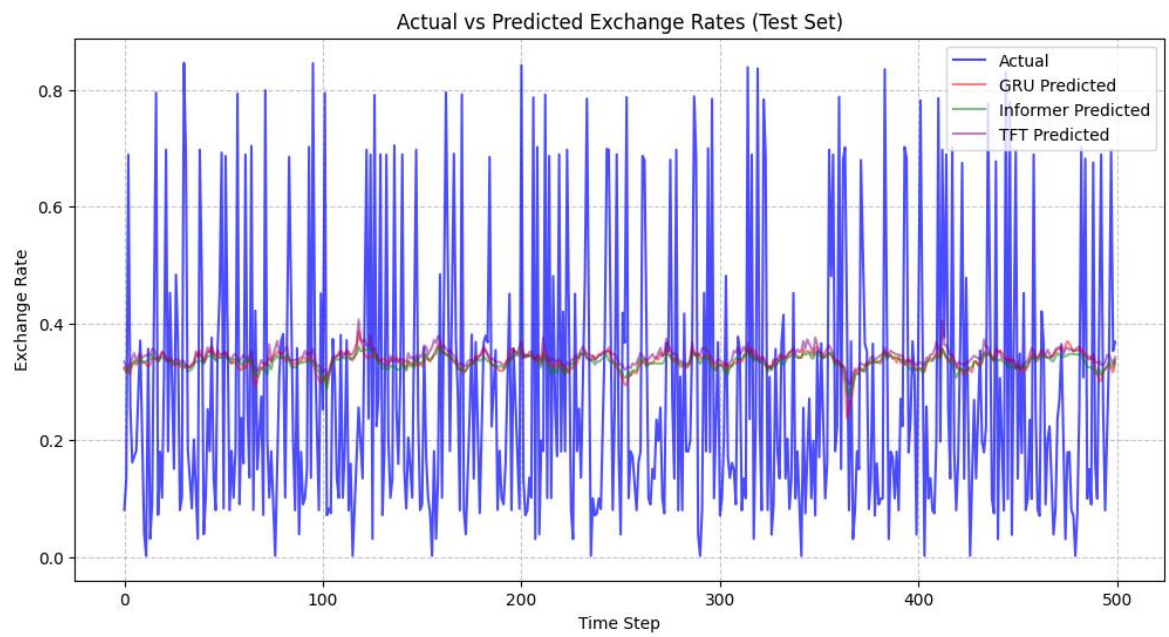


Figure A2.4 Actual vs Predicted Exchange Rates (Test Set)

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